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REVIEW

Ectomycorrhizal fungi in tropical Africa: hidden diversity, spatial turnover, and functional effects on tree growth

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Abstract

Tropical Africa encompasses several biodiversity hotspots, yet knowledge of ectomycorrhizal (EcM) fungi, key organisms for forest ecosystem functioning, remains fragmented. This study provides the first comprehensive synthesis of EcM fungi across the region, aiming to (i) establish a species checklist as a basis for estimating gamma diversity, (ii) analyze spatial turnover across countries, and (iii) assess the functional effects of EcM fungi on tree seedling growth. We conducted a systematic literature search following PRISMA guidelines, retaining 158 publications published between 1927 and 2025. Species occurrence records were extracted from publications and complemented with data from GlobalFungi and GBIF databases. Statistical analyses in R were used to estimate species richness (Chao2, Jackknife 1 and Jackknife 2), analyze spatial turnover (Jaccard dissimilarity, Mantel correlogram, Generalized Dissimilarity Modelling), and assess the overall effect on tree seedling growth (cross-classified random-effects meta-analysis). We recorded 711 EcM fungal species from 59 genera and 20 families. *Russula* (169 species), *Amanita* (81), and *Lactifluus* (71) were the most represented. Estimated richness reached 1,226 species, indicating substantial undocumented diversity. Community composition showed extremely high heterogeneity among countries (95% dissimilarity), and increased turnover over broader geographic distances ($r = -0.13$, $p < 0.05$), suggesting dispersal limitation. Meta-analysis revealed that EcM fungi increased tree seedling growth by an average of 6% per month ($p < 0.05$), with *Scleroderma verrucosum* showing particularly strong positive effects. These findings indicate that EcM fungi are unevenly distributed across tropical Africa. They refine our understanding of diversity patterns and highlight the need for targeted sampling and integrating native EcM fungi into forest restoration.

Keywords Ectomycorrhizal fungi · Biogeography · Conservation ecology · Distance decay · Meta-analysis

Introduction

Tropical Africa encompasses several biodiversity hotspots, including the West African forests, the Eastern Arc and coastal forests of Kenya and Tanzania, and Madagascar (Myers et al. 2000; Küper et al. 2004). Within the region, forest ecosystems are highly heterogeneous and form a complex mosaic of biomes (White 1983; Droissart et al. 2018). West Africa is characterized by a gradient from humid Guinean forests to savanna woodlands, whereas Central Africa is dominated by the Congo Basin rainforest, comprising evergreen and semi-deciduous forest types (White 1983). East Africa is characterized by a mosaic of savanna woodlands (including miombo), montane forests, and coastal forests, while Southern Africa exhibits a diverse vegetation mosaic including extensive miombo woodlands

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in its northern parts and Afromontane forests (White 1983; Wilmé et al. 2012).

These ecosystems host a rich fungal diversity that sustains ecosystem functioning and provides essential services to human societies (Yorou et al. 2014; Tedersoo et al. 2014; Niego et al. 2023; Adjimoti et al. 2025). Among them, ectomycorrhizal (EcM) fungi play a key ecological role by forming mutualistic associations with dominant tree species such as *Afzelia* Sm., *Brachystegia* Benth., *Gilbertiodendron* J. Léonard, *Isoberlinia* (Harms) Craib & Stapf, *Julbernardia* Pellegr., and *Uapaca* Baill. (Högberg and Piearce 1986; Sanon et al. 1997; Onguene and Kuyper 2001; Ducousso et al. 2008; Bâ et al. 2011, 2012; Corrales et al. 2018). These associations enhance nutrient and water acquisition for trees and increase their resistance to environmental stress (Onguene et al. 2013; Bahram et al. 2014; Zheng et al. 2014; van der Heijden et al. 2015; Pellitier et al. 2021). In particular, West-Central Africa has recently been identified as one of the global centers of endemic soil EcM fungi (Tedersoo et al. 2022).

Understanding EcM fungal diversity in tropical Africa is important for forest conservation and for better apprehending ecosystem functioning. Despite their ecological importance, EcM fungi face increasing threats in the region from anthropogenic pressures, particularly deforestation, with a net annual forest loss of 3.9 million hectares between 2010 and 2020 (FAO 2020). Such degradation is likely to cause long-term biodiversity loss (Malhi et al. 2013; Bauters et al. 2021; FAO 2024) and may irreversibly alter EcM fungal communities (see, for example, Rodriguez-Ramos et al. 2021 and Teteli et al. 2024). Effective conservation and restoration strategies therefore require a comprehensive understanding of the diversity, spatial distribution, and ecological functioning of EcM fungi across the region.

While molecular and integrative approaches have significantly advanced our knowledge of EcM fungi in the region by revealing numerous species new to science (De Kesel et al. 2016; Hyde et al. 2019; Codjia et al. 2020; Aïgnon et al. 2021a, b; Badou et al. 2022), existing information remains geographically, methodologically, and taxonomically fragmented. Consequently, a comprehensive understanding of EcM fungal diversity at the scale of tropical Africa (i.e., gamma diversity) is still lacking. For instance, Piepenbring et al. (2020) reported 247 EcM species in West Africa, representing 5.1% of the total fungal diversity recorded for the subregion. Comparable syntheses are unavailable for other African subregions, preventing assessments of continental-scale diversity (Pinkert et al. 2022).

Understanding gamma diversity requires more than merely compiling species checklist. It also depends on evaluating two complementary components of biodiversity: within-region richness (alpha diversity) and among-region compositional differences (beta diversity). The latter can be assessed through patterns of spatial turnover and distance-decay relationship, which describe the decline in community similarity with increasing geographic distance. Historically, microbial distributions were interpreted under the ''everything is everywhere'' hypothesis (Baas-Becking 1934), which assumes that microorganisms are not limited by dispersal. However, increasing evidence suggests that dispersal limitation play important roles in shaping fungal communities, particularly in host-associated systems (Lilleskov et al. 2004; Bahram et al. 2015; Glassman et al. 2017). Despite these advances, fungal biogeography remains strongly biased toward temperate regions (Peay et al. 2010; Tedersoo et al. 2012, 2014; Davison et al. 2018; Delhaye et al. 2024), leaving such spatial patterns largely unexplored across tropical Africa.

From a conservation and restoration perspective, documenting taxonomic diversity and spatial distribution is only the first step; it is equally essential to evaluate the functional significance of this biodiversity. Specifically, understanding how EcM fungi affect host tree growth will provide important insights into their role in ecosystem resilience. Experimental studies conducted across tropical Africa have shown that EcM fungi can enhance tree seedling growth, particularly under nutrient-poor or degraded conditions (Delwaulle et al. 1982; Bâ et al. 1999; Diédhiou et al. 2005; Diagne et al. 2013; Hounsouvo et al. 2022). Nevertheless, these studies remain scattered, and no meta-analysis has yet evaluated whether functional responses to EcM inoculation show consistent trends at a continental-scale.

To address these interconnected knowledge gaps, this study integrates published literature on EcM fungi in tropical Africa, complemented by occurrence records from GlobalFungi and GBIF databases, within a single coherent conceptual framework. Specifically, we aim to: (i) establish a species checklist as a basis for estimating gamma diversity and revealing hidden diversity; (ii) analyze spatial turnover and distance-decay of EcM fungal communities across the region; and (iii) assess their functional effects on tree seedling growth. By linking diversity, spatial distribution, and functional performance, this study aims to advance our understanding of tropical African EcM fungal communities and provide evidence-based insights for forest conservation.

Methods

Geographic focus

This review covers 38 tropical African countries across four major geographic subregions (Fig. 1). The study area encompasses a wide diversity of vegetation types, including

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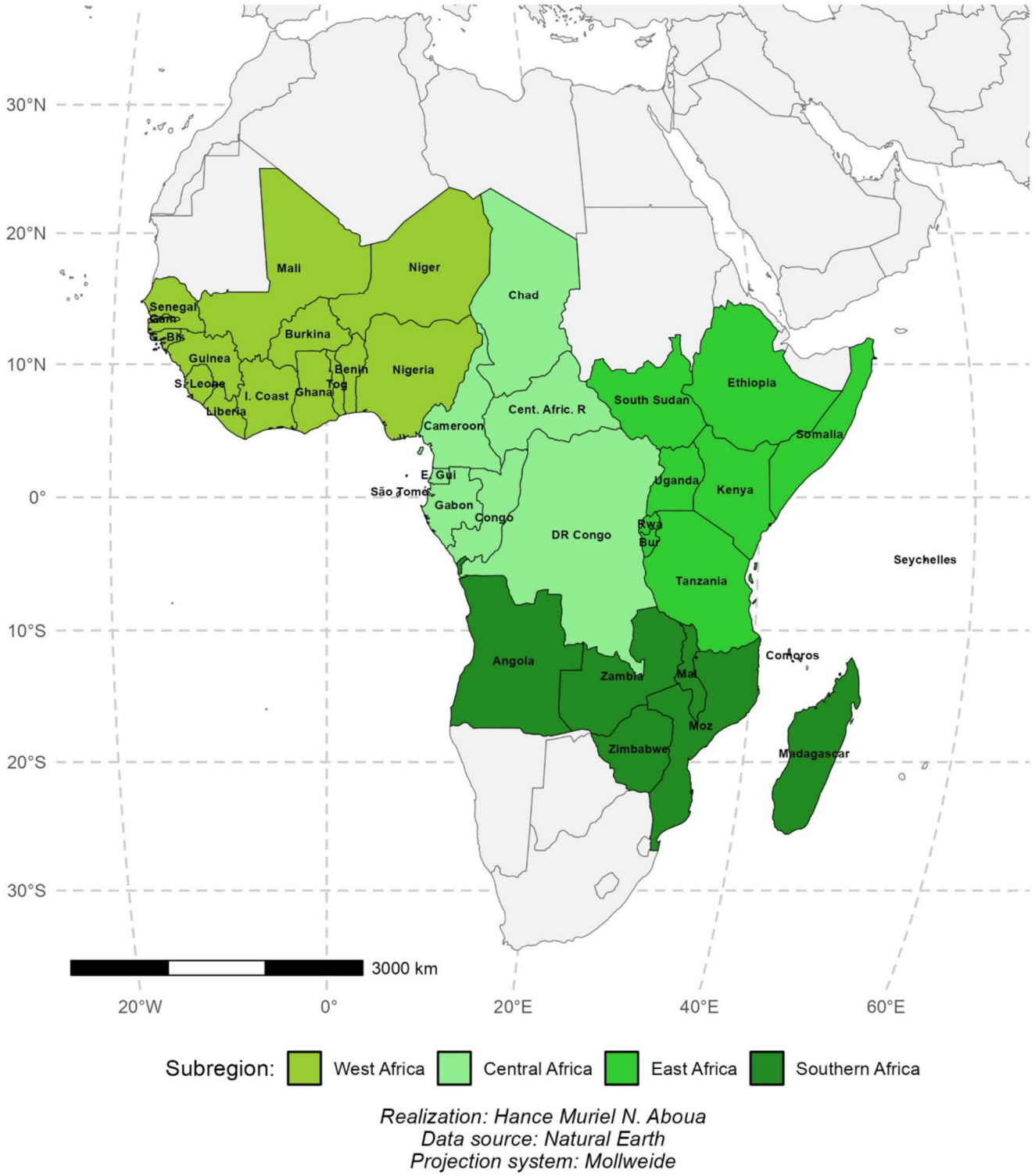


Fig. 1 Map of Africa showing the boundaries of the included tropical countries, organized by subregion. *Bur = Burundi; Burkina = Burkina Faso; Cent. Afric. R = Central African Republic; DR Congo = Democratic Republic of the Congo; E. Gui = Equatorial Guinea; G.-Bis = Guinea Bissau; Gam = Gambia; I. Coast = Ivory Coast; Mal = Malawi; Moz = Mozambique; Rwa = Rwanda; São Tomé = São Tomé and Principe; S. Leone = Sierra Leone; Tog = Togo*

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dense humid rainforests, deciduous and dry forests, riparian forests, as well as woodlands and wooded savannas, which support a wide diversity of EcM trees (Table 1).

The climate across tropical Africa is predominantly tropical (Köppen-Geiger type A), but exhibits strong spatial heterogeneity in rainfall and temperature seasonality (Peel et al. 2007). Climate data were processed using a geospatial workflow implemented in Google Earth Engine (https://earthengine.google.com/; Gorelick et al. 2017), which enabled spatial aggregation and extraction of climatic variables at the subregional scale. Mean annual temperature was extracted from WorldClim BIO1, while annual precipitation was extracted from WorldClim BIO12. Country polygons were merged to define subregional geometries, and climate variables were summarized only within forested areas, based on the ESA WorldCover 2020 dataset (tree cover class 10). Summary statistics include mean values and spatial variability expressed as minimum and maximum values. Detailed climatic information for each subregion is provided in Table 1.

Table 1 Vegetation, EcM tree, and climatic characteristics of tropical African subregions

Subregion	Major vegetation types	Dominant EcM tree genera	Mean temperature (range)	Mean precipitation (range)
West Africa	Guinean forests, gallery forests, Sahelian savannas	*Afzelia*, *Berlinia*, *Isoberlinia*, *Monotes*, *Uapaca*	26.0 °C (16.2–30.5 °C)	1,914 mm (36−4,402 mm)
Central Africa	Congo Basin dense forests, montane forests, semi-deciduous forests	*Gilbertiodendron*, *Tetraberlinia*, *Uapaca*	24.2 °C (9.1–29.9 °C)	1,818 mm (8−4,143 mm)
East Africa	Miombo woodlands, montane forests, coastal forests	*Brachystegia*, *Isoberlinia*, *Julbernardia*	22.4 °C (5.4–30.7 °C)	1,232 mm (14−2,625 mm)
Southern Africa	Miombo woodlands, Afromontane forests, mosaic of dry and humid forests	*Asteropeia*, *Sarcolaena*, *Uapaca* (endemic to Madagascar)	21.9 °C (11.9–28.0 °C)	1,126 mm (46−3,354 mm)

Data on major vegetation types were compiled from White (1983), Poorter et al. (2004), CARPE (2005), and Wilmé et al. (2012). Information on dominant EcM tree genera was compiled from Bâ et al. (2012) and Corrales et al. (2018)

The primary threats to biodiversity across tropical Africa include deforestation, habitat fragmentation, shifting cultivation, charcoal production, and overexploitation of natural resources (Gumbo et al. 2010; FAO, 2020). These pressures are driven by a combination of demographic growth, agricultural expansion, and increasing demand for ecosystem services (Chidumayo et al. 1991; Gumbo et al. 2010; Assédé et al. 2020; FAO, 2020, 2024).

Systematic literature search strategy

To build a comprehensive dataset of EcM fungi in tropical Africa, we developed a rigorous and transparent approach, aligned with established best practices in evidence synthesis, including PRISMA (Page et al. 2021). The process began with the identification of key search terms derived from an initial set of ten relevant seed publications (see Online Resource 1 in Supporting Information), selected to represent the breadth of existing research in line with the study's main objective. These terms, encompassing the population (e.g., EcM, ectomycorrhizal) and the outcomes of interest (e.g., diversity, distribution, growth), were combined into Boolean search strings (Table 2) and iteratively refined to ensure capture of the seed publications.

The final search strings were applied to the bibliographic databases Scopus (https://www.scopus.com) and Web of Science (https://www.webofknowledge.com), covering all publications up to September 12, 2025. In Scopus, searches were conducted within the field ''Article title, Abstract, and Keywords'', while in Web of Science, the ''Topic'' field was used. The searches resulted in a total of 2,542 records (854 publications from Scopus and 1,688 from Web of Science; Fig. 2). A preliminary test in CAB Abstracts (https://www.cabidigitallibrary.org/product/ca) did not yield any unique relevant records and was therefore not retained in the final search strategy.

Article selection and screening process

The records identified from the databases were combined into a single dataset, and duplicates were removed using the '*revtools*' package (Westgate 2019) in R software (v.4.1.3, R Core Team). After removing duplicates, 2,018 unique records remained (Fig. 2). These records then underwent a screening process in three sequential steps: titles, abstracts, and full texts. Publication eligibility was assessed against predefined criteria covering publication topic, geographic area, language, and type (Table 2). From the initial database

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Table 2 Search strategy and eligibility criteria

Search string used in the bibliographic databases		
	Search string	Access date
Scopus; Web of Science	(EcM OR "ectomycor*" OR "mycor*" OR "wild mushroom*" OR "macromyc*") AND ("Africa*" OR Afrique OR Madagascar OR "tropical forest*") AND ("diversit*" OR distribution OR "new species" OR "inoculat*" OR growth)	September 12, 2025
Eligibility criteria applied to the publications		
	Inclusion criteria	Exclusion criteria
Publication topic	Studies documenting the diversity or geographic distribution of EcM fungi in plant formations, with fungal taxa identified at the binomial level and the country of occurrence clearly specified; or reporting controlled EcM inoculation experiments on woody plant species, including a non-inoculated control group, plant performance metrics (height or total biomass), and the number of replicates in both inoculated and control treatments to allow calculation of response ratios.	Studies not involving EcM fungi, based solely on local knowledge, or lacking species-level taxonomic confirmation; as well as controlled EcM inoculation experiments involving non-woody plants or agricultural crops, using multiple EcM fungal strains per treatment, or lacking a non-inoculated control group or quantitative plant performance data required to compute effect sizes.
Geographic area	Tropical African countries (see Fig. 1).	Countries outside tropical Africa.
Language of publication	English (the dominant language in publications) or French (official language of many tropical African countries).	Studies not published in English or French.
Publication type	Original articles published in peer-reviewed journals and available in full text.	Review articles, theses, books, conference proceedings, or any non-peer-reviewed documents.

search, 75 full-text articles were assessed for eligibility (Fig. 2). Four of these were excluded (Online Resource 1). Following the completion of this rigorous screening, 71 articles were retained as eligible for inclusion.

The 71 retained articles were used as the relevant evidence base to perform a complementary citation search. This was achieved using the Citationchaser tool (https://estech.shinyapps.io/citationchaser/; Haddaway et al. 2022) to conduct backward and forward citation tracking, which initially yielded 2,724 unique publications (Fig. 2). Subsequent screening of the titles of these publications identified 93 potentially relevant articles. These articles were subjected to two-step screening process, first abstracts, then full texts, and to the same eligibility criteria as the initial records. After this screening, six articles were excluded (Online Resource 1), resulting in the inclusion of 87 articles. This multi-source approach allowed for a comprehensive bibliographic coverage of the primary literature, while confirming the high specificity of citation-based search in identifying relevant studies (Wohlin et al. 2022).

The included articles comprise 146 articles on EcM fungal diversity and distribution and 12 articles reporting controlled inoculation experiments. To ensure consistency and minimize selection bias, 10% of the screening decisions made by one reviewer were randomly selected and independently checked by another reviewer, who was blinded to the original decisions. Discrepancies between the reviewers were resolved through discussion until consensus was reached. Deduplication and title and abstract screening were carried out using annotated R scripts (see Online Resource 6), while full text screening was conducted manually.

Data collection and compilation

Bibliographic data were extracted for each selected publication, including authors, year of publication, title, journal, and the country where the study was conducted. For the three publications (Bâ et al. 2002; Founoune et al. 2002; Diédhiou et al. 2004) in which the study area was unspecified, we contacted one of the authors to obtain this information. For studies conducted in the Democratic Republic of the Congo (DR Congo, formerly Zaire), the only case in which data were originally recorded under a former name, the country name was standardized to ensure consistency. Studies conducted in at least two tropical African countries are considered multi-country.

A species-level database of EcM fungi was compiled from all relevant publications, listing each species once in alphabetical order. For each species in the checklist, taxonomic information (genus, species, family, order, and phylum), geographic location, reference sources, and when available, the associated host tree and ecosystem were recorded. The checklist integrates data from fruit body surveys, soil DNA analyses, and EcM root samples. To ensure taxonomic reliability, taxa identified only to the genus level and uncertain identifications (e.g., marked as sp., cf., or aff.) were excluded, while infraspecific taxa were considered at the species level. Names were checked on November 3, 2025 for validity and synonymy against Index Fungorum (https://www.indexfungorum.org/) and Mycobank (https://www.mycobank.org/), and only names currently accepted in at least one of these databases were retained. Additional occurrence records for species reported in the reviewed publications,

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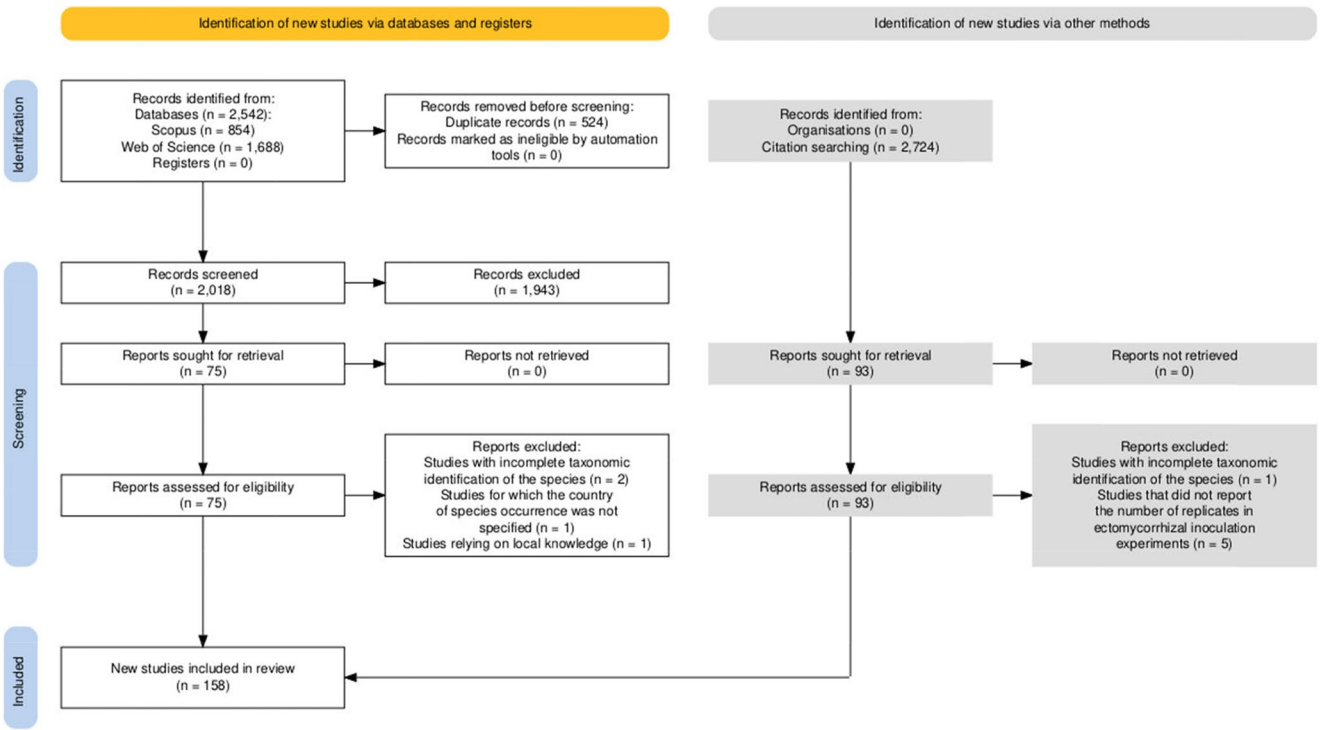


Fig. 2 PRISMA flow diagram summarizing the process of publications selection

as well as for additional species, were retrieved from the GlobalFungi database (https://globalfungi.com/) and the GBIF database (https://www.gbif.org/). Additional species were included only when their genera were recognized as EcM based on the reviewed publications and FungalTraits 1.2 (Põlme et al. 2020).

Data on inoculation experiments were also compiled for the meta-analysis, including the fungal and tree species tested, experimental conditions, and duration. For each study, we extracted quantitative data on plant growth responses, including means for inoculated and control treatments, sample sizes, and measures of variability (standard deviations or standard errors when available), which were used to compute effect sizes.

Data analyses

All analyses and figure generation were conducted in R, with reproducibility ensured through provision of annotated scripts in the Online Resource 6.

Analyses of temporal trends and geographic coverage of publications

Temporal trends in publications were analyzed using Spearman's rank correlation on the number of publications per five-year period, with numeric ranks assigned to the periods. Results were visualized with bar plots. Disparities in publications distribution among countries were quantified using the Gini index (Gini 1914; Weiner and Solbrig 1984) with the '*ineq*' package (Zeileis 2014), and visualized with Lorenz curves (Sadras and Bongiovanni 2004). Statistical significance was assessed through 9,999 random permutations. Geographic coverage of publications was summarized with a hierarchical sunburst chart.

Analyses of diversity and species richness estimation

Species richness (α-diversity) was calculated as the total number of recorded species per phylum, order, family, genus, subregion, and country, and presented as bar plots. To account for potentially unrecorded species, species richness across tropical Africa was estimated using the non-parametric Chao2 estimator, followed by the Jackknife 1 and Jackknife 2 estimators in the '*vegan*' package (Oksanen et al. 2022). Analyses were based on a species-by-publication presence-absence matrix, where each publication reporting at least one fungal occurrence was treated as an independent sampling unit. In this framework, the number of publications was used as a proxy for sampling effort. To assess sampling completeness across tropical Africa, a species accumulation curve was constructed using publications as pooling units, applying a random resampling procedure with 9,999 permutations. Confidence intervals were derived from permutation-based variation among randomized accumulation paths. Chao2 and Jackknife estimates

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were computed from the same presence-absence matrix and subsequently overlaid on the accumulation curve. For visualization purposes, asymptotic richness estimates were projected along the sampling gradient to illustrate potential undocumented diversity under increased publication effort.

Analyses of compositional dissimilarity and spatial turnover

Cosmopolitan species were defined as those present in at least one country of each subregion. Species rarity was approximated using a species-specific geographic restriction score, calculated as the inverse of the number of countries in which a species occurs, such that species documented from a single country received the highest score. While this approach does not fully align with Rabinowitz's multidimensional rarity framework (Rabinowitz 1981), it provides a simple measure of geographic restriction and helps identify taxa for which individual countries may bear particular conservation responsibility (Schmeller et al. 2008).

Pairwise species dissimilarity (β-diversity) among countries was quantified using Jaccard distances from a binary presence-absence matrix, implemented in the '*vegan*' package. Mean pairwise dissimilarity was calculated across all country combinations (excluding self-comparisons) to provide a continental-scale baseline. To account for uneven sampling effort, a sensitivity analysis was conducted using only the five best-documented countries in our dataset. Variation in community composition among subregions was tested with PERMANOVA (9,999 permutations) based on a Jaccard distance matrix (Anderson 2001), followed by PERMDISP (Anderson 2006) to test homogeneity of multivariate dispersions. Post-hoc pairwise PERMANOVA tests were performed with Benjamini-Hochberg correction to identify specific subregional differences. Compositional patterns were visualized using principal coordinates analysis (PCoA) based on the Jaccard dissimilarity. To assess robustness of between-subregion variation, PERMANOVA was repeated excluding the country with the highest species richness, which could disproportionately influence variation and obscure patterns among the remaining countries.

Spatial structure was evaluated using a Mantel test and a Mantel correlogram (9,999 permutations) based on Jaccard dissimilarity and Vincenty geographic distances between country centroids (used as spatial proxies due to inconsistent georeferencing in the literature). Distance classes in the correlogram were optimized to ensure a reliable number of country pairs per class. Generalized Dissimilarity Modelling (GDM) in the '*gdm*' package (Manion et al. 2018) was used to quantify non-linear patterns of community turnover along geographic gradients and to estimate the proportion of deviance explained by distance.

Meta-analysis of inoculation experiments

For inoculation experiments, we conducted a meta-analysis to evaluate the effects of EcM fungal inoculation on tree growth across tropical Africa, and identify the taxon associated with the strongest positive growth response. The growth improvement of inoculated seedlings relative to controls was quantified using the response ratio (*R*), calculated by dividing the mean value of the growth variable in the inoculated treatment ($\bar{X}i$) by the mean value of the same variable in the control treatment ($\bar{X}c$).

To ensure temporal comparability across experiments of different durations, the Monthly Improvement Ratio (*MIR*) was computed as:

$$MIR = R^{\frac{1}{d}} = \left(\frac{\overline{X}i}{\overline{X}c}\right)^{\frac{1}{d}}$$

where *d* is the duration of the experiment in months.

Effect sizes were expressed as the log-response ratio (*LRR* = *ln(MIR)* ; Hedges et al. 1999), where *ln* is the natural logarithm.

Following the ''missing-cases'' method of Nakagawa et al. (2023), the coefficient of variation (*CVi* for inoculated treatments and *CVc* for control treatments) was calculated from the two studies (Duponnois et al. 2005; Diagne et al. 2013) reporting standard errors (*SE*):

$$CVi = \frac{SE \times \sqrt{n}}{\overline{X}i}, CVc = \frac{SE \times \sqrt{n}}{\overline{X}c}$$

Weighted mean *CV*s were used to estimate sampling variance (*vi*) for all studies, including those lacking *SE* or *SD* (standard deviation):

$$\nu_i = \frac{CVi_2}{ni} \times \frac{CVc^2}{nc}$$

where *ni* and *nc* are the sample sizes of the inoculated treatment and control, respectively.

Meta-analysis was conducted in the '*metafor*' package (Viechtbauer 2010) using a cross-classified random-effects model fitted via Restricted Maximum Likelihood (REML) to account for non-independence and hierarchical structure:

$$LRR = \mu + \mu_s + \mu_c + \mu_t + \mu_f + \varepsilon_{sctf}$$

where μ is the overall mean; μ_s, μ_c, μ_t, μ_f are the random effects for study, country, host tree species, and EcM fungal species, respectively; and ε_{sctf} is the residual error of each effect size, with variance specified by the diagonal of the variance-covariance matrix $V = \text{diag}(vi)$.

REML estimates and 95% confidence intervals were reported; variance components (σ^2) were extracted for each random effect, I^2 statistics quantified heterogeneity by hierarchical level (Higgins and Thompson 2002), and Cochran's *Q* assessed residual heterogeneity. Robustness was evaluated via two sensitivity analyses: a Leave-One-Out analysis, iteratively excluding one observation at a time, and

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exclusion of the three *LRR*s with the largest *vi* to assess the influence of high-variance observations. Finally, a forest plot was generated, showing individual *LRR*, study labels, and the pooled estimate with 95% confidence intervals.

Results

Temporal trends and geographic inequalities in published research on EcM fungi in tropical Africa

After a rigorous selection process, a total of 158 publications were retained (see Online Resource 3), published between 1927 and 2025. The analysis of temporal trend by 5-year periods shows an overall increase in the number of publications ($r = 0.95$, $p < 0.001$), with a marked acceleration beginning in the late 1990s and a peak occurring between 2016 and 2020 (Fig. 3a).

The geographic coverage of publications is characterized by strong inequality, as evidenced by a high Gini index (Gini = 0.49, $p < 0.001$; Fig. 3b). West Africa accounts for the highest number of publications (32.28%, $n = 51$) whereas East Africa accounts for the lowest (10.13%, $n = 16$; Fig. 3c). Benin (13.29%, $n = 21$), Madagascar (8.86%, $n = 14$), and Cameroon (7.59%, $n = 12$) are the most represented countries. Multi-country publications account for 24.05%.

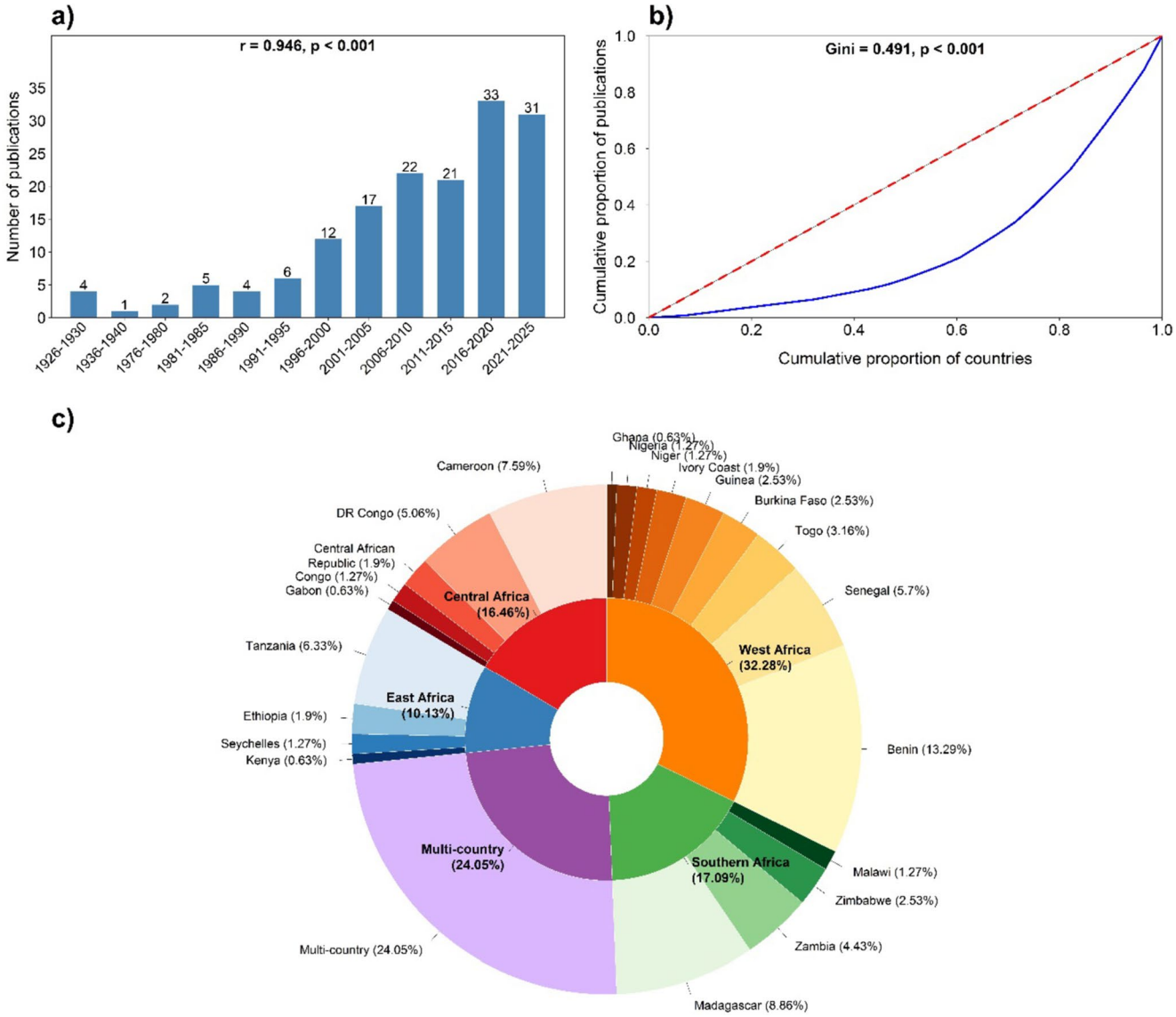


Fig. 3 Published research output. **a)** Barplot of the evolution over time of the number of publications on EcM fungi in tropical Africa, grouped by five years periods; **b)** Lorentz curve showing the cumulative proportion of publications by the cumulative proportion of countries; **c)** Hierarchical sunburst chart showing subregional distribution of publications. *''Multi-country'' refers to publications in which research was conducted across multiple countries, at regional or continental scales*

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Known diversity and estimated species richness (γ-diversity)

A total of 711 EcM fungal species have been recorded, with 2,476 country-level occurrences (checklist in Online Resource 4). They belong to 59 genera, 20 families, and 10 orders, and distributed across 2 phyla. The phylum *Ascomycota* accounts for 0.6% of the taxa, comprising two orders, whereas *Basidiomycota* represents 99.4% and includes eight orders (Fig. 4a). The order *Russulales* is the most represented, with 288 species, whereas the order *Pezizales* is the least represented, with only one species. At the family level, the *Russulaceae* is by far the most represented with 288 species, followed by the *Boletaceae* (110 species) and the *Amanitaceae* (82 species) (Fig. 4b). At the genus level, the most represented genera are *Russula* (169 species), *Amanita* (81 species), and *Lactifluus* (71 species) (Fig. 4c). Central Africa harbors the highest known species richness (436 species), while East Africa harbors the lowest (273 species) (Fig. 4d). Known species richness per country varies, ranging from 1 species recorded in Angola, Chad and Guinea-Bissau to 354 species in the DR Congo (Fig. 4d). Apart from the DR Congo, Cameroon (214 species), Zambia (198), Madagascar (178), and Benin (171) have the highest species richness.

The non-parametric Chao2 estimator suggests an estimated species richness of 1,052 species across tropical Africa, whereas the Jackknife 1 and Jackknife 2 estimators yield 1,056 and 1,226 species, respectively (Fig. 5). These estimations indicate that many species remain undocumented. The species accumulation curve does not reach an asymptote, indicating that current sampling remains incomplete.

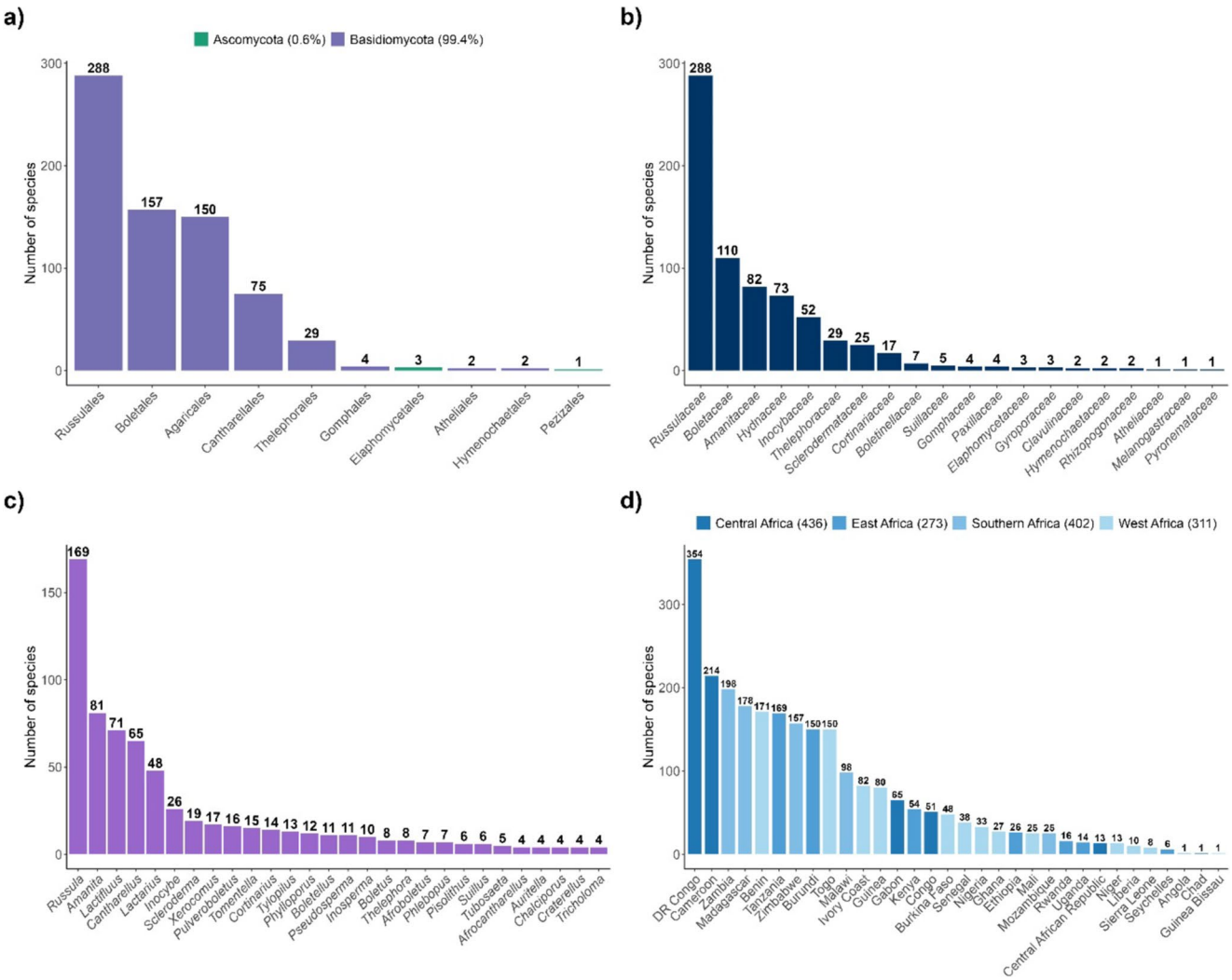


Fig. 4 Number of EcM fungal species. **(a)** By phylum and by order; **(b)** By family; **(c)** By genus (top 25 most species-rich genera, including ties); **(d)** By subregion and by country (only countries with recorded species are shown)

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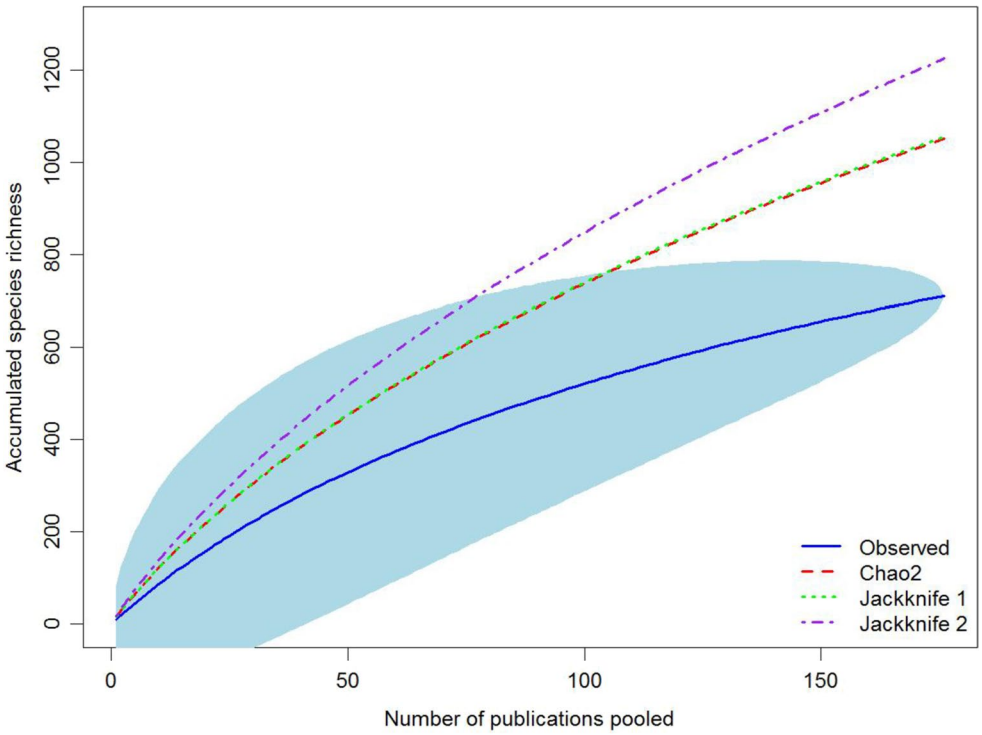


Fig. 5 Cumulative and estimated species richness of EcM fungi as a function of sampling effort (number of publications pooled)

Species composition and patterns of spatial turnover

Among the recorded species, 114 species (16.03%) were found to be cosmopolitan, occurring across all subregions (Table S1 in Online Resource 2). *Coltricia cinnamomea* (Jacq.) Murrill, *Lactifluus gymnocarpoides* (Verbeken) Verbeken, *Lactifluus gymnocarpus* (R. Heim ex Singer) Verbeken, *Lactifluus pelliculatus* (Beeli) Buyck, *Phlebopus sudanicus* (Har. & Pat.) Heinem., and *Tubosaeta brunneosetosa* (Singer) Horak were the most cosmopolitan species, occurring in at least 15 countries (Table S2). Other widely distributed species included *Amanita loosei* Beeli, *Amanita masasiensis* Härk. & Saarim., *Amanita rubescens Pers.*, *Amanita strobilaceovolvata* Beeli, *Cantharellus densifolius* Heinem., *Porphyrellus niger* Heinem. & Gooss.-Font., *Russula annulata* R. Heim, and *Russula congoana* Pat., each recorded in more than 12 countries. A total of 265 species (37.27%) were recorded in only a single country of tropical Africa, suggesting geographically restricted distributions, although this pattern may partly reflect uneven sampling effort. These spatially restricted species belong predominantly to the genera *Amanita*, *Cantharellus*, *and Russula*, with Central Africa hosting the highest numbers (Table S3).

Mean Jaccard dissimilarity among countries was extremely high (95%), revealing extreme spatial heterogeneity in species composition and confirming that most species are restricted to one or a few countries. Even when considering only the five best-documented countries (Benin, Cameroon, DR Congo, Madagascar, and Zambia), dissimilarity remained high (77.57%), suggesting that this pattern is not an artifact of uneven sampling. Across all countries, PERMANOVA revealed a significant but modest effect of subregion on fungal community composition ($R^2 = 0.132$, $F = 1.419$, $p < 0.001$), indicating that subregional differences explain 13.2% of total variation. Multiple-testing corrections in the pairwise comparisons identify significant differences between West Africa and both East Africa and Central Africa ($p < 0.05$, see Table S4), whereas no other pairwise comparisons remain significant after adjustment. PCoA similarly showed broadly dispersed country assemblages, with the first two axes accounting for 18.3% of the variation (Fig. 6). PERMDISP confirmed no significant differences in multivariate dispersion among subregions ($p = 0.68$). The subregional effect remained significant after excluding the country with the highest species richness ($R^2 = 0.134$, $F = 1.397$, $p < 0.001$; Fig. S1), demonstrating robustness to potential sampling bias.

The Mantel test revealed a significant correlation between geographic distance and community dissimilarity ($r = 0.19$, $p = 0.004$), indicating decreasing similarity with increasing distance. The Mantel correlogram revealed a significant scale-dependent spatial structure in EcM fungal communities (Fig. 7). At the shortest geographic distance class (centered at 554 km), a significant positive spatial autocorrelation was detected ($r = 0.28$, $p < 0.001$), indicating higher similarity among geographically close communities. At an intermediate distance class (centered at 2,916 km), a significant

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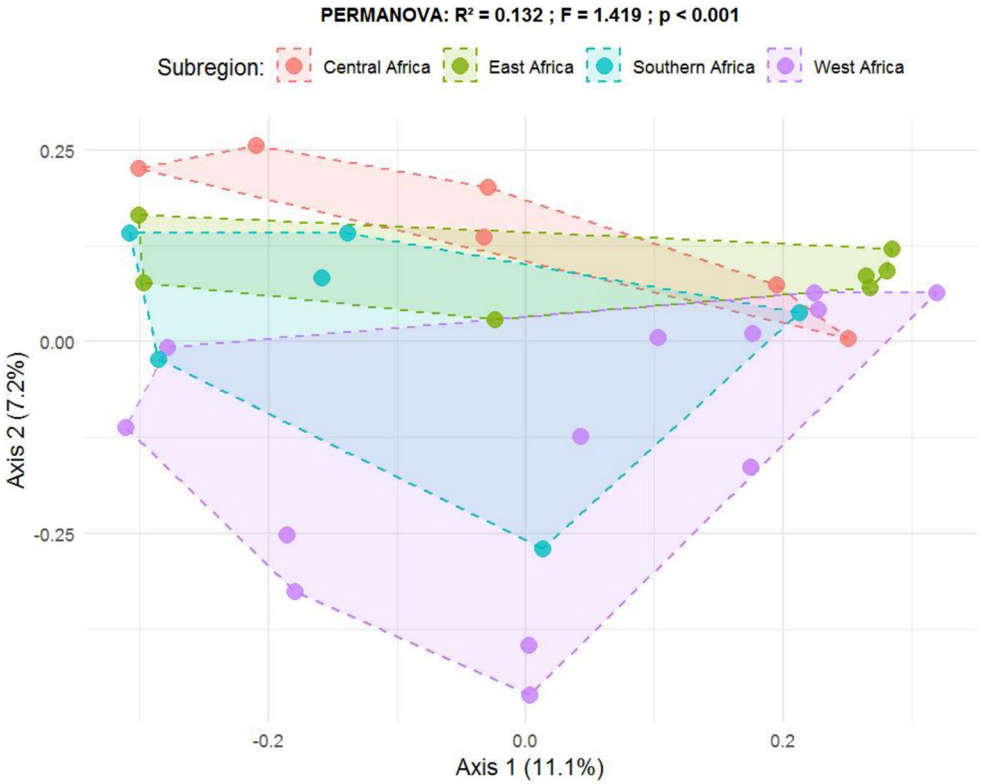


Fig. 6 PCoA plot of EcM fungal community dissimilarity among countries. *Subregions are delimited by convex hulls to highlight compositional clustering*

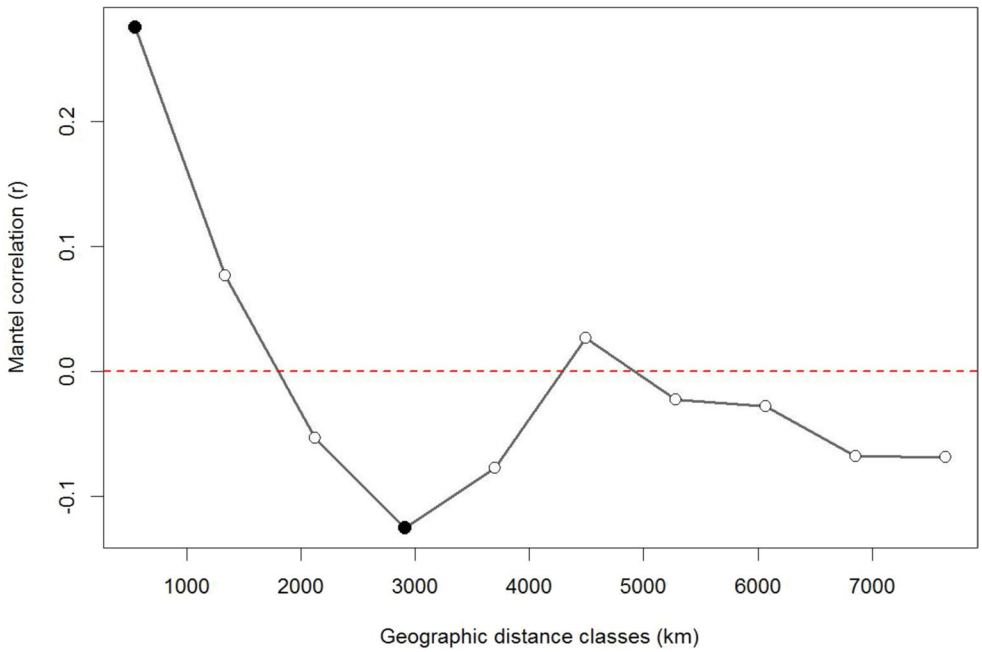


Fig. 7 Mantel correlogram showing the spatial autocorrelation of EcM fungal community dissimilarity across geographic distance. *Black-filled circles represent statistically significant correlations ($p < 0.05$), whereas white circles indicate non-significant values*

negative autocorrelation was detected (r = -0.13, $p < 0.05$), suggesting increased compositional turnover across regional scales. At larger distances, no significant autocorrelation was detected ($p > 0.05$), indicating that spatial structure weakened at broader continental scales. GDM regression analysis supported these findings, with geographic distance explaining 7.34% of the total model deviance (Null deviance = 45.06; Model deviance = 41.76). The high dissimilarity values (0.80–0.98; Fig. 8a) combined with the high intercept (Intercept = 1.70) indicate a strong baseline level of compositional turnover. Furthermore, the shape of the geographic spline (Fig. 8b) reveals a highly non-linear pattern, characterized by a steep increase in species turnover at short geographic distances, followed by a plateau-like phase at larger distances.

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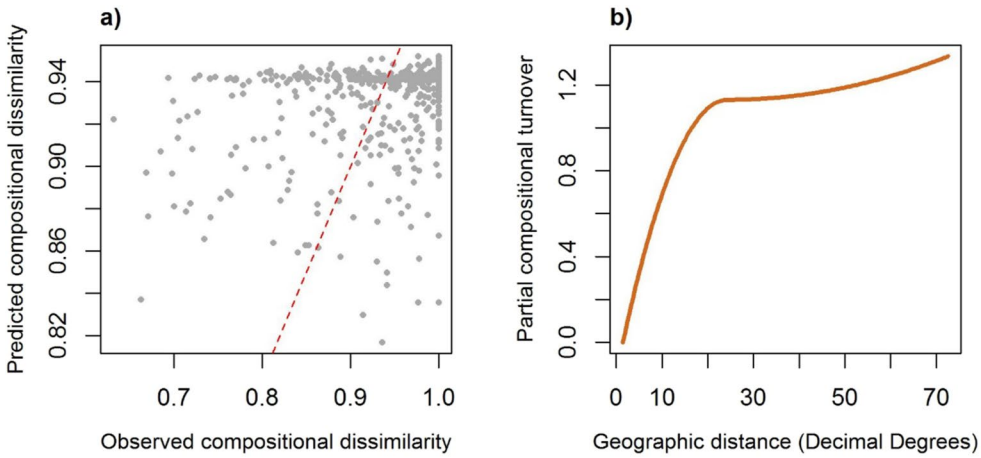


Fig. 8 GDM of EcM fungal community turnover in relation to geographic distance. **(a)** Relationship between observed and predicted compositional dissimilarity; **(b)** Regression spline showing the contribution of geographic distance to compositional turnover

Functional performance of EcM fungi on tree seedling growth

The meta-analysis showed a significant positive effect of EcM fungi on tree growth ($p<0.05$). Within the subset of species currently tested, inoculation resulted in an average growth increase of 6% per month (mean effect estimate = 0.06; Fig. 9). *Scleroderma verrucosum* (Bull.) Pers. showed the strongest effect (Fig. 9, Online Resource 5), yet it was recorded in only eleven countries of the 38 tropical African countries, corresponding to a geographic occurrence coverage of 28.94%.

Variance component estimates indicated that most residual heterogeneity was attributable to tree species ($\sigma^2 = 0.002$), while variation due to publication, country, or fungal species was negligible ($\sigma^2 < 10^{-10}$, Table 3). The total heterogeneity (I^2) was 19.90%, entirely explained by differences among tree species. The Q statistic ($Q=38.18$, $df=47$, $p>0.05$) indicates that there is no significant heterogeneity among the studies.

Leave-One-Out analysis showed that no single study disproportionately influenced the effect of EcM fungi on tree growth (mean effect estimate ranging from 0.050 to 0.061, $p<0.05$ in all cases; Table S5). Similarly, excluding the three studies with the largest variance yielded a similar estimate (mean effect = 0.059, $p<0.05$; Table S6).

Discussion

The extent of hidden diversity of EcM fungi in tropical Africa

This study establishes the first quantitative baseline of EcM fungal diversity in tropical Africa. However, because knowledge of EcM fungal diversity remains incomplete, richness estimators likely only partially reflect true species diversity in the region. The estimated richness of 1,226 species is consistent with Verbeken and Buyck (2002), who suggested that diversity might exceed 800 species. Even this most optimistic estimate represents only 5% of the approximately 25,000 EcM fungal species estimated worldwide (Rinaldi et al. 2008; van Galen et al. 2025), highlighting substantial underrepresentation of tropical African EcM fungi. Comparisons with other tropical regions further support this pattern. For example, Corrales et al. (2022), using metabarcoding datasets from the GlobalFungi database, reported 713 species hypotheses in Southeast Asia compared to only 344 in the Afrotropics. This contrast is not explained by EcM host tree diversity, as Africa harbors a higher number of confirmed EcM tree species (120) than Asia (59; Corrales et al. 2018), suggesting that observed differences in fungal richness primarily reflect unequal sampling effort.

The low percentage of *Ascomycota* compared to *Basidiomycota* observed in this study is consistent with global trends in EcM fungal diversity (Rinaldi et al. 2008). Nevertheless, these results reveal substantial hidden diversity and highlight the need for targeted sampling in underrepresented countries and poorly sampled habitats, including montane, riparian, dry, humid, and semi-deciduous forests. The 311 species reported here for West Africa, compared with the 247 species documented by Piepenbring et al. (2020), likely reflects increased sampling effort in recent years.

The present checklist integrates data from fruit body surveys (accounting for most studies), soil DNA analyses (e.g., Tedersoo et al. 2022), and EcM root assessments (e.g., Bauman et al. 2016), while prioritizing taxonomically validated records. However, substantial heterogeneity in sampling approaches limits comparability across studies, as detection probability and taxonomic resolution differ among methods. Fruit body surveys may underestimate diversity by missing non-fruiting taxa, whereas DNA-based approaches can recover a broader range of taxa, including cryptic lineages. Conversely, DNA-based approaches may also involve

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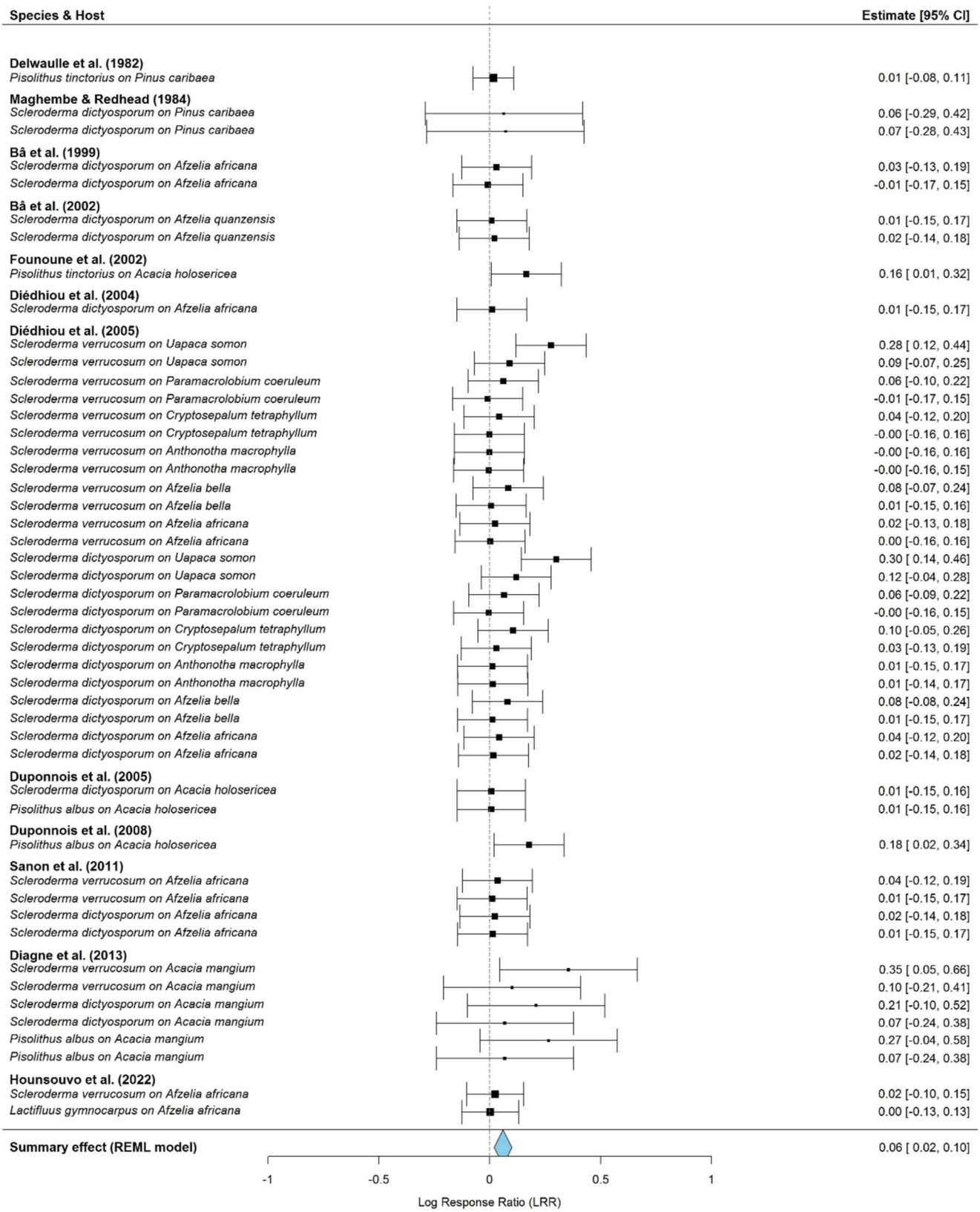


Fig. 9 Species-specific effects and overall effect of EcM fungi on tree growth. *An LRR greater than 0 indicates growth higher than the control, while an LRR less than 0 indicates growth lower than the control*

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Table 3 REML estimates of model parameters and variance components from a cross-classified random-effects meta-analysis

	Parameter	Value
Global model	Number of effect size	48
	Mean effect estimate	0.06
	SE	0.020
	z-value	2.888
	p-value	0.003
Variance components and associated statistics	Number of publication	12
	Publication σ^2	1.92e-12
	Publication I^2	1.58e-08
	Number of country	5
	Country σ^2	5.61e-12
	Country I^2	4.63e-08
	Number of tree species	11
	Tree species σ^2	2.41e-03
	Tree species I^2	1.99e+01
	Number of EcM fungal species	5
	EcM fungal species σ^2	1.35e-14
	EcM fungal species I^2	1.11e-10
	Total I^2	1.99e+01

limitations such as possible misidentification or insufficient taxonomic resolution at the species level. Beyond methodological differences, the uneven spatial distribution of available records also introduces limitations. Countries were used as operational spatial units because most records are reported at the national scale, although they do not represent ecologically homogeneous or equally sampled units. Several countries (Comoros, Equatorial Guinea, Gambia, São Tomé and Príncipe, Somalia, and South Sudan) remain data-deficient, reflecting sampling gaps rather than true absences. Furthermore, while the exclusion of ambiguous taxonomic records (e.g., genus-level assignments, cf., aff.) reduces the risk of inflated richness, it may also lead to conservative estimates.

Despite these limitations, the checklist synthesizes dispersed knowledge from multiple sources and provides a harmonized reference for EcM fungi in tropical Africa. It represents a valuable foundation for conservation prioritization, monitoring species' responses to climate change, biogeographic comparisons, and future macroecological analyses (Piepenbring et al. 2020; Pinkert et al. 2022). However, finer-scale, spatially explicit approaches will be essential for improving our understanding of diversity patterns in future studies.

Ecological insights and conservation implications of spatial turnover patterns

Our analyses reveal that 16.03% of species occur across all subregions of tropical Africa, suggesting broad geographic distributions. By contrast, more than 30% of species occur in only one country. This pattern includes several *Cantharellus*, *Lactifluus*, *and Russula* species known exclusively from tropical Africa, which are likely endemic (Verbeken and Buyck 2002; Tedersoo et al. 2010, 2022). Together, these results highlight strong geographic restriction among many taxa across the continent and identify potentially range-restricted species that may deserve conservation attention. Despite this high degree of geographic restriction, the ecological traits and life-history characteristics that may explain the pronounced spatial restriction observed in many taxa remain poorly understood. Nevertheless, caution is warranted when interpreting rarity based solely on country-level distributions, as species occurring only in large countries may occupy extensive geographic ranges, whereas those recorded from small countries may appear more geographically restricted than they actually are. Central Africa harbor the highest number of restricted species, consistent with Tedersoo et al. (2022), who identified the subregion as one of the major global centers of endemic soil fungi, including EcM taxa. However, tropical Africa faces anthropogenic pressures such as deforestation, agricultural expansion, and logging, all of which directly affect fungal communities. For example, Teteli et al. (2024) reported significant declines in EcM fungal diversity following forest conversion to farmland, suggesting that some highly localized and potentially endemic lineages may already have gone extinct before even being documented.

The high dissimilarity observed between countries indicates that EcM fungal communities are strongly spatially structured across tropical Africa, with pronounced compositional turnover. This pattern supports the findings of Tedersoo et al. (2011) for certain sites in Africa, and is consistent with the spatial structuring of fungal communities reported in temperate and boreal forests (Delhaye et al. 2024). Community composition also differed among subregions. Although the first two PCoA axes explain a limited fraction of the total variation, they nonetheless reveal non-random spatial organization in community composition. Significant positive spatial autocorrelation at short distances indicates spatial structuring of EcM fungal communities. Conversely, negative correlations at larger distances reflect increasing compositional turnover. This scale-dependent pattern aligns with the framework proposed by Odriozola et al. (2024) in Europe, where dispersal processes tend to dominate community assembly at local scales, whereas environmental filtering becomes more important at broader regional scales. In tropical African EcM fungi, the sharp increase in dissimilarity at short geographic distances likely reflects limited dispersal, constraining spore movement even among relatively proximate regions. In contrast, the observed tendency toward stabilization of dissimilarity at larger distances may reflect the influence of environmental heterogeneity.

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Taken together, these patterns support strong spatial structuring of EcM fungal diversity across tropical Africa at both species and community levels. While environmental predictors were not explicitly included due to data limitations, some studies highlight the potential role of environmental factors in shaping community turnover (Tedersoo et al. 2011, 2012, 2014; Bauman et al. 2016; Vanié-Léabo et al. 2017; Badou et al. 2021). In particular, host tree diversity has been shown to predict EcM fungal diversity (Gao et al. 2013, 2014), suggesting that variation in the number and composition of dominant hosts may help maintain fungal diversity across tropical forests. Likewise, spatially structured soil properties (Wang et al. 2002) and lineage-specific distributions of dominant EcM taxa (Bahram et al. 2013) may contribute to the observed distance-decay patterns. Species-specific study has also shown that climatic variables can strongly influence EcM fungal distribution (Aboua et al. 2025), further supporting the role of climate as a key structuring factor. Overall, our findings contradict the "everything is everywhere" hypothesis, supporting a model in which both dispersal constraints and spatially structured environmental variation contribute to EcM community composition. Future studies integrating quantitative environmental and ecological variables across multiple spatial scales will be essential to explain the residual variation in EcM fungal turnover.

Host tree identity as a major source of heterogeneity in tree growth responses and a link between spatial distribution and functional importance

Through meta-analysis, this study provides the most comprehensive quantitative synthesis to date of the effects of EcM fungi on tree growth in tropical Africa, offering an illustrative example of the functional implications of their spatial distribution. Across the studies included, inoculated seedlings exhibited an average monthly growth increase of 6%, with *Scleroderma verrucosum* emerging as the most effective species. This finding aligns with observations from the Neotropics, where species of the genus *Scleroderma* Pers. are widely used as reliable inoculants (Baeza-Guzmán et al. 2025). Sensitivity analyses confirmed the robustness of this positive growth response. An important finding emerging from the integration of distribution patterns and functional analyses is that *S. verrucosum*, the taxon showing the strongest positive effect on tree growth, has been recorded in only a few countries across tropical Africa. This spatial restriction suggests that key ecological functions may depend on taxa with limited distributions. Consequently, forest degradation could disproportionately affect such mutualists, particularly if their restricted distributions and limited dispersal capacities reduce their ability to recolonize disturbed habitats naturally.

Despite these promising results, only five EcM fungal species have been evaluated in inoculation trials, highlighting major gaps in our understanding of fungal functional diversity. If species with restricted distributions, such as *S. verrucosum* can exhibit a strong effect on tree growth, many other rare or geographically limited EcM fungi may also possess important but currently unrecognized ecological functions. Therefore, functional assessments should be expanded to include a broader range of EcM taxa, particularly poorly studied species. Importantly, functional effects are not consistent across EcM species. Some taxa may provide greater growth benefits to host trees, whereas others may have neutral or even negative effects. For example, in an experiment conducted in Europe, Rincón et al. (2007) found that inoculation with *Suillus collinitus* (Fr.) Kuntze significantly improved the survival of *Pinus halepensis* Mill. seedlings, whereas inoculation with *Rhizopogon roseolus* (Corda) Th. Fr. caused significant seedling mortality. These contrasting responses highlight the importance of fungal species identity in determining inoculation outcomes. However, the fact that most of the studies included in our review were conducted under greenhouse conditions may limit the transferability of the findings to natural forest ecosystems. Under field conditions, climatic variability and soil fertility gradients can strongly influence the effects of EcM fungi on seedling performance (Anthony 2025). Nevertheless, evidence from natural forests suggests that the ecological importance of EcM fungi extends beyond controlled experiments, with effects also documented at the forest scale (Anthony et al. 2022).

Variance component estimates revealed that host tree identity explained most of the observed heterogeneity in growth responses, whereas differences among fungal species, countries, or publications contributed comparatively little. This pattern suggests that variation among host tree species may be an important source of variability in EcM growth responses. However, because these factors were included as random effects rather than tested as explicit moderators, the results should be interpreted as a partitioning of unexplained heterogeneity rather than evidence of causal drivers. Moreover, the limited representation of host diversity and geographic coverage in the current literature may restrict the generality of these patterns across tropical Africa. Therefore, additional studies spanning a broader range of countries and host tree species are needed to strengthen inference and support future continent-wide syntheses.

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Conclusion

This study provides the first comprehensive quantitative baseline of EcM fungal diversity in tropical Africa, addressing a geographic gap in fungal biogeography and providing meta-analytic evidence of the effects of EcM fungi on tree growth. By consolidating data from one of the least-studied regions, we reveal exceptionally diverse but largely undocumented fungal communities and highlight substantial hidden diversity. These results underscore the need for continued systematic field surveys.

Our analyses indicate strong spatial structuring of EcM fungal communities, with pronounced compositional turnover across countries. While dispersal limitation shapes community assembly across broader geographic distances, continental patterns suggest that environmental factors may also be important. We further show that EcM fungi can enhance tree seedling growth by an average of 6% per month, underscoring their ecological importance for forest regeneration and productivity. Notably, the species associated with the strongest positive growth response, *S. verrucosum*, was restricted to a few countries across tropical Africa. This pattern suggests a potential mismatch between geographic distribution and ecological importance, whereby highly beneficial EcM fungi may have relatively limited ranges.

Overall, this study advances our understanding of the diversity, biogeography, and functional importance of EcM fungi in tropical Africa. It provides a foundation for future continent-wide analyses, conservation planning, and evidence-based forest restoration, while raising questions about the mechanisms driving both spatial turnover and functional effects.

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Author contributions H.M.N.A., R.D., A.D.M.T.H., R.D.H., and N.S.Y. conceived the study. H.M.N.A., A.D.M.T.H., G.B.H., and N.S.Y. developed the methodology. H.M.N.A. conducted the literature search, screening, and data curation; performed the data analysis and visualization; and led the writing of the first draft of the manuscript. G.B.H. independently checked a random 10% subset of the screening decisions to ensure consistency. R.D., S.A.B., K.I.T., A.-A.B., and H.L.A. verified the datasets compiled from bibliographic sources. N.S.Y. supervised the study. All authors revised the manuscript and approved the submitted version.

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Data availability All data generated or analyzed during this study are fully included in the Supporting Information.

Code availability The R scripts used in this study for deduplication of records, title and abstract screening, data analysis and figure generation are provided in the Supporting Information.

Declarations

Competing interests The authors declare no competing interests.

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